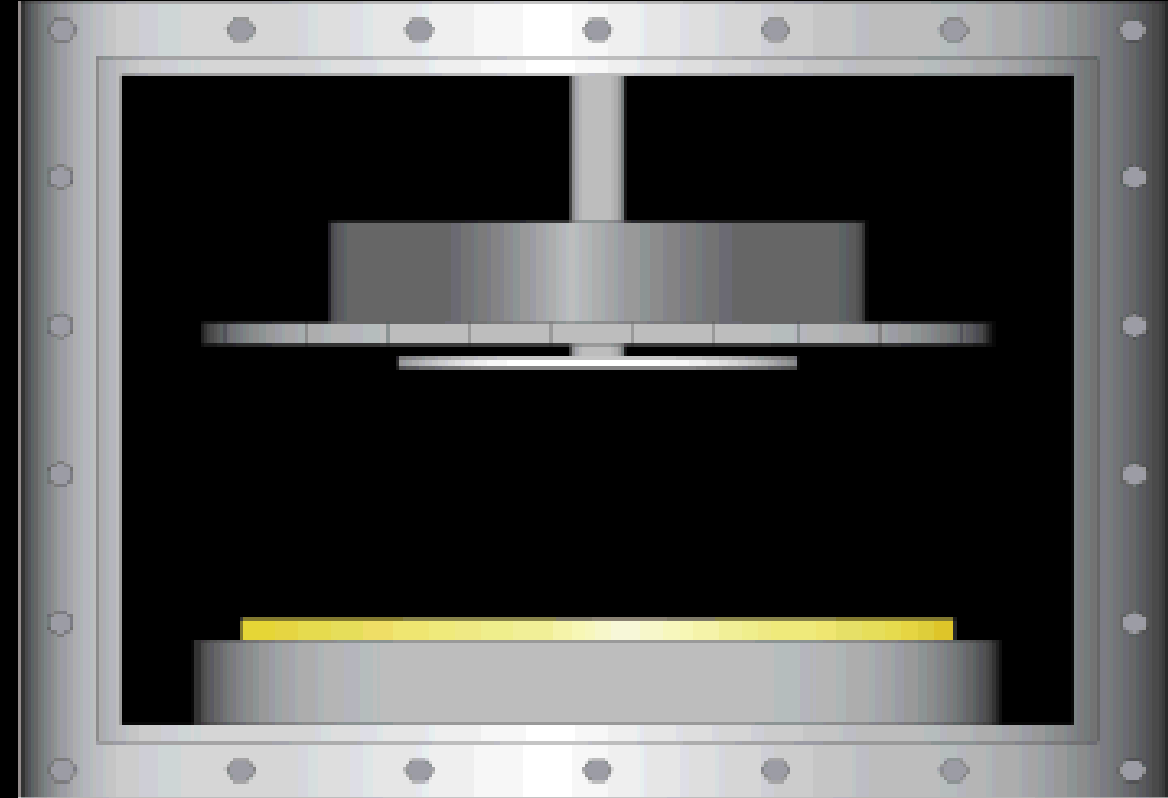


A photograph of a sputtering process inside a vacuum chamber. A central substrate holder is mounted on a rotating stage, with several small, dark, irregularly shaped sputtered targets or pieces of material resting on it. The chamber is surrounded by various mechanical components and wiring. The image is overlaid with a vibrant, abstract graphic consisting of flowing, translucent bands in shades of red, orange, yellow, green, and blue, creating a dynamic, futuristic feel.

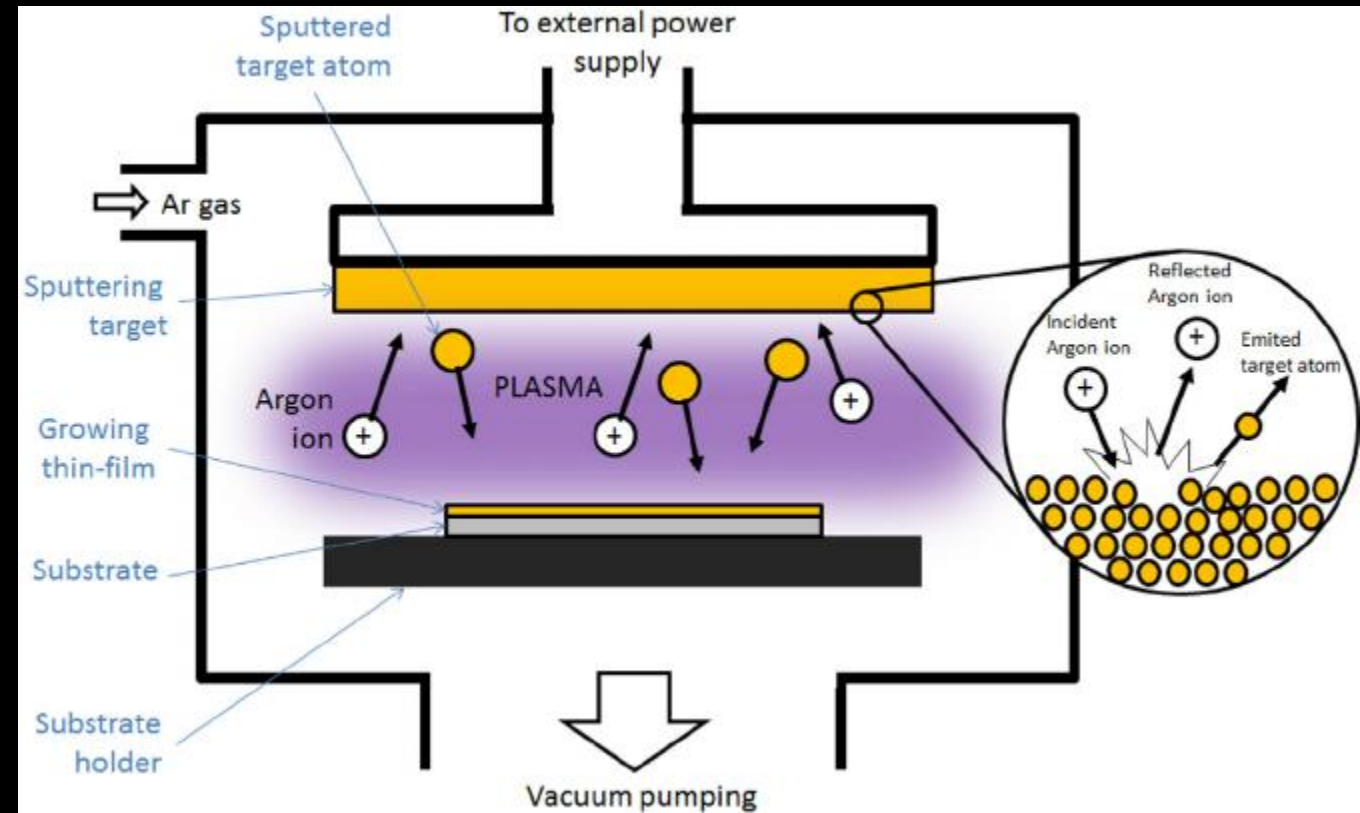
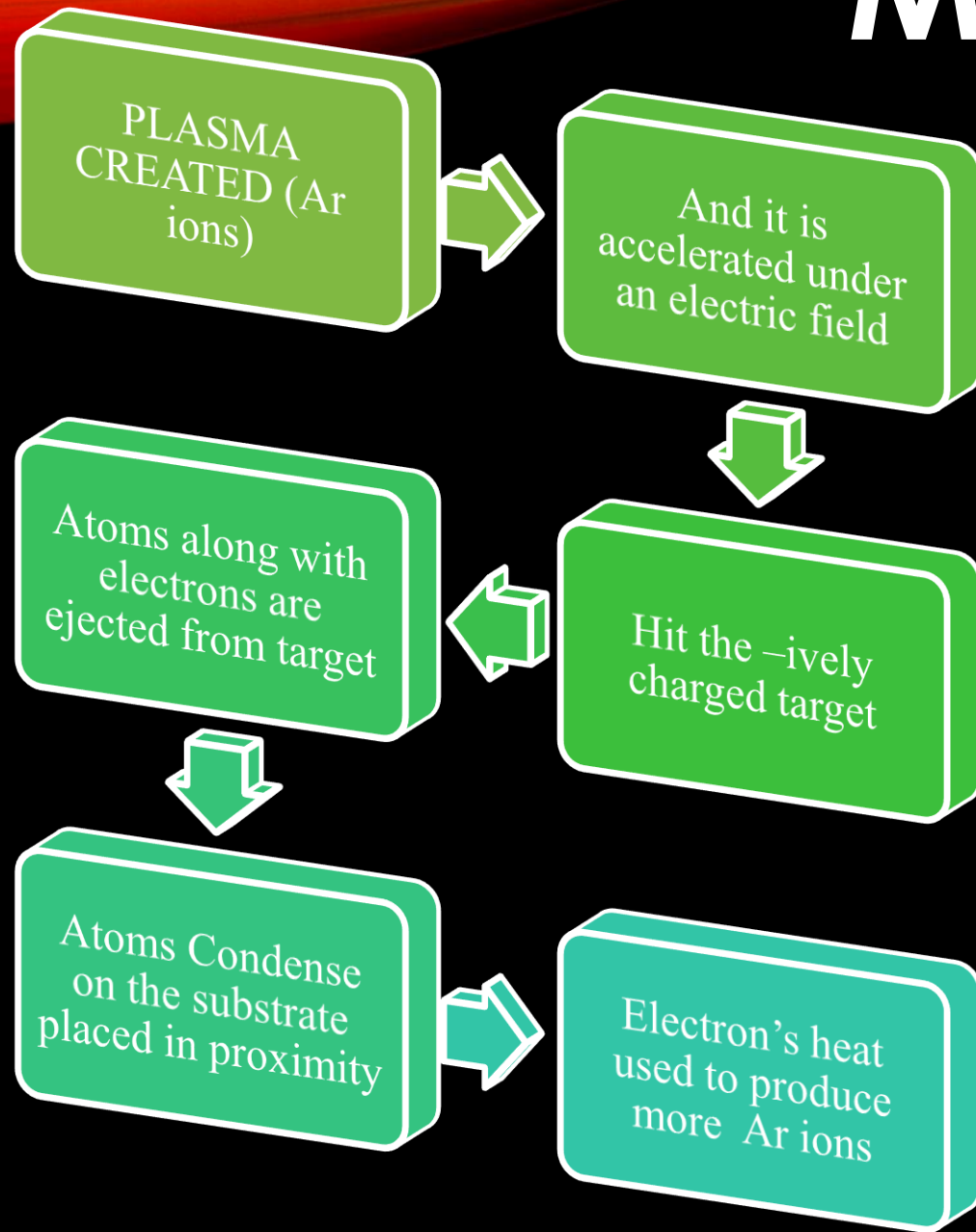
SPUTTERING

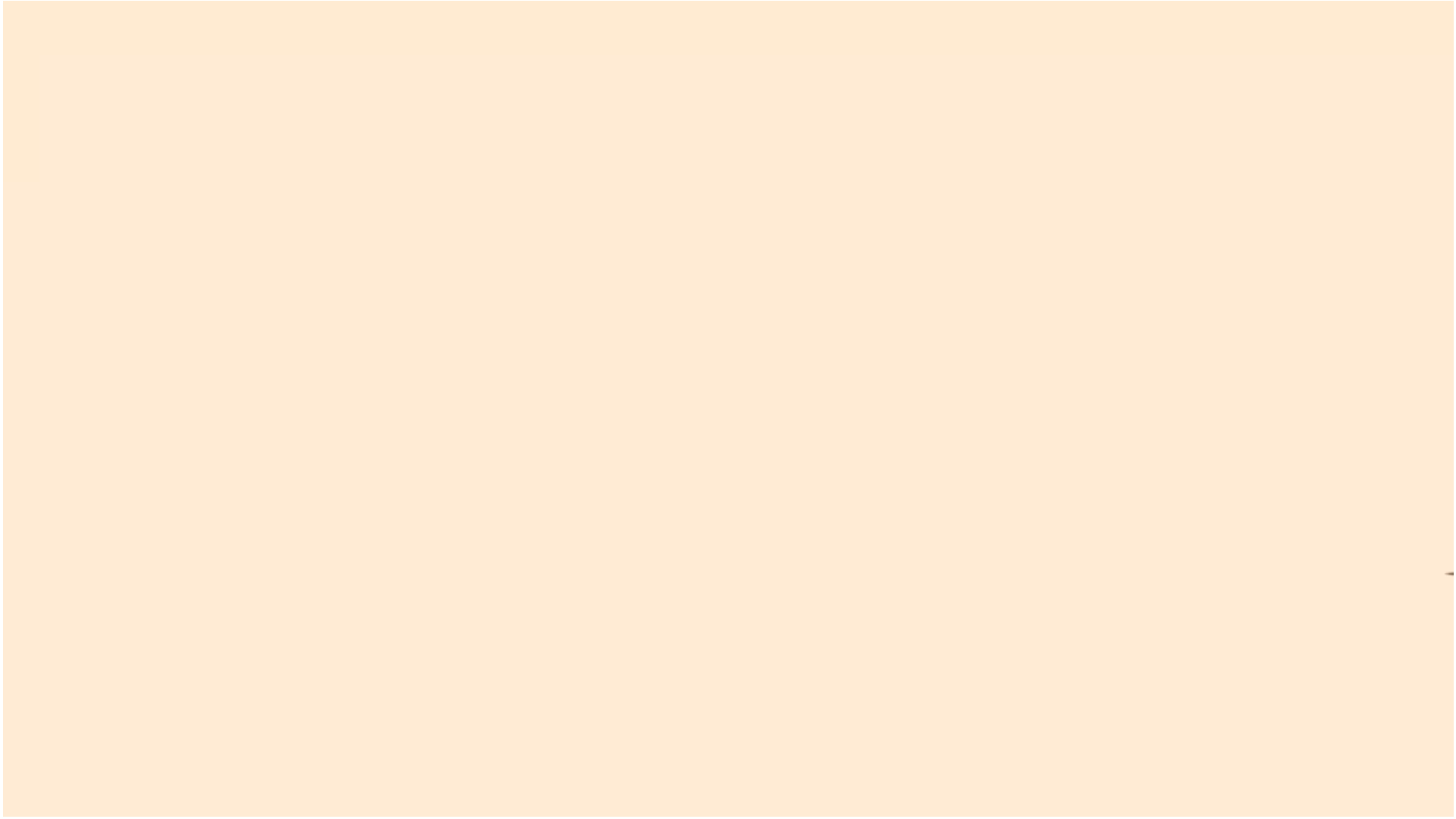
INTRODUCTION TO SPUTTERING

- **Sputtering:** A **non-thermal vaporization process** where surface atoms are ejected by momentum transfer from energetic atomic or molecular bombardment.
- Typically **glow discharge** or **ion beams** to generate incident ions.
- Heavy ions like argon bombard the target, ejecting atoms into the gas phase.
- Operates in a **vacuum** with inert gas at 10^{-2} to 10^{-1} torr and 500–1000 V potential.
- **Applications:** Sputter etching (removal) and sputter deposition (re-deposition onto a substrate).



METHODOLOGY





ION GENERATION IN SPUTTERING

- Plasma discharge: Electrons from the cathode ionize inert gas atoms, creating a glow discharge.
- Positively charged ions accelerate toward the cathode, causing sputtering.
- Ion beams: High-beam current sources focus ions directly onto the target.
- Sputtering yield: Fixed number of atoms ejected per ion.
- Magnetron systems use magnetic fields to enhance plasma density near the target.

TARGET SURFACE INTERACTIONS

- Positive ions cause multiple interactions:
 - Liberation of neutral/ionized atoms.
 - Backscattering and secondary electron emission.
 - Desorption of gas atoms.
- Internal target effects:
 - Collisional cascades and point defect creation.
 - Local heating, amorphization, and implantation.

Reflection Ions and Neutrals

Incident Ion

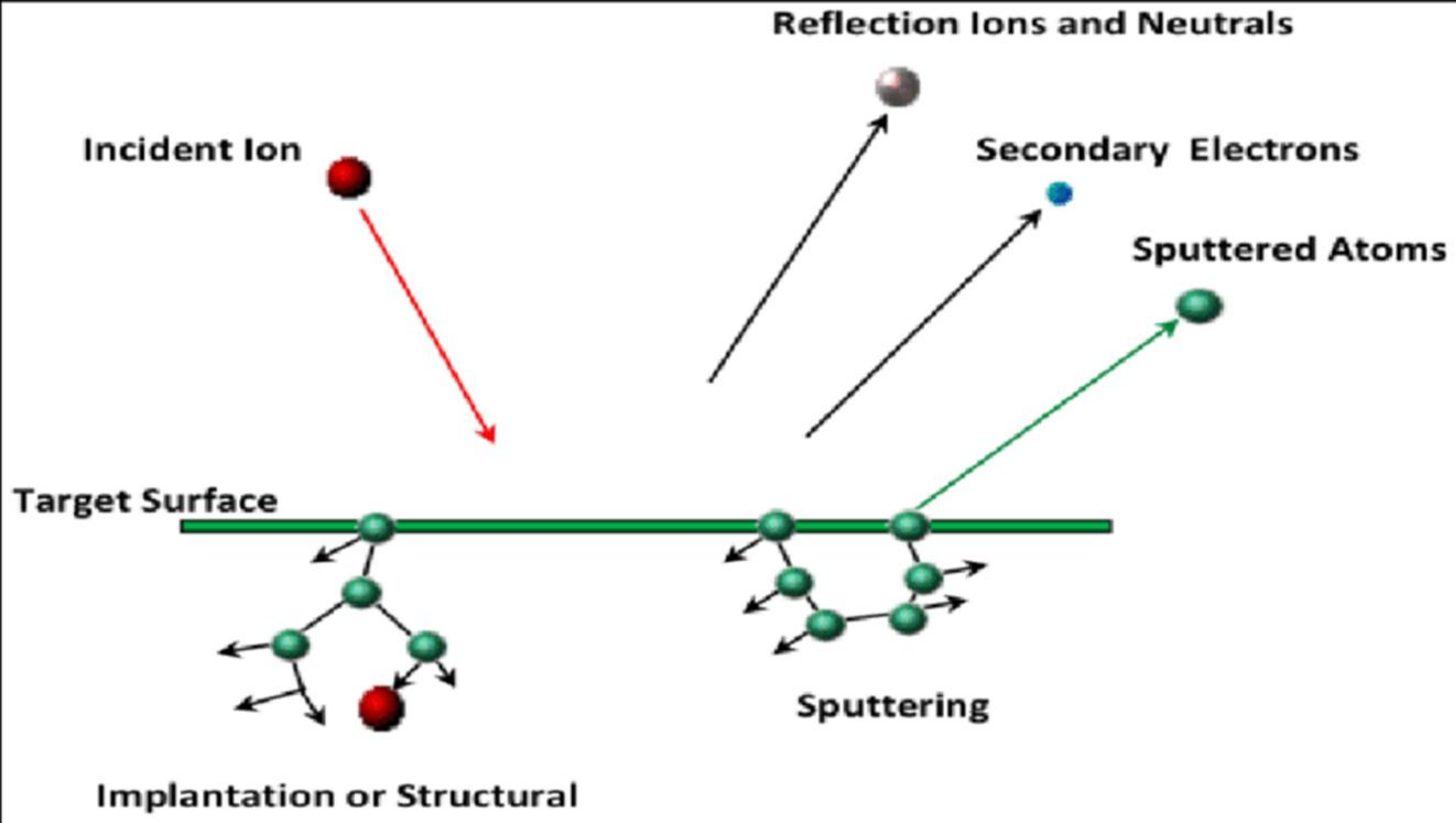
Secondary Electrons

Sputtered Atoms

Target Surface

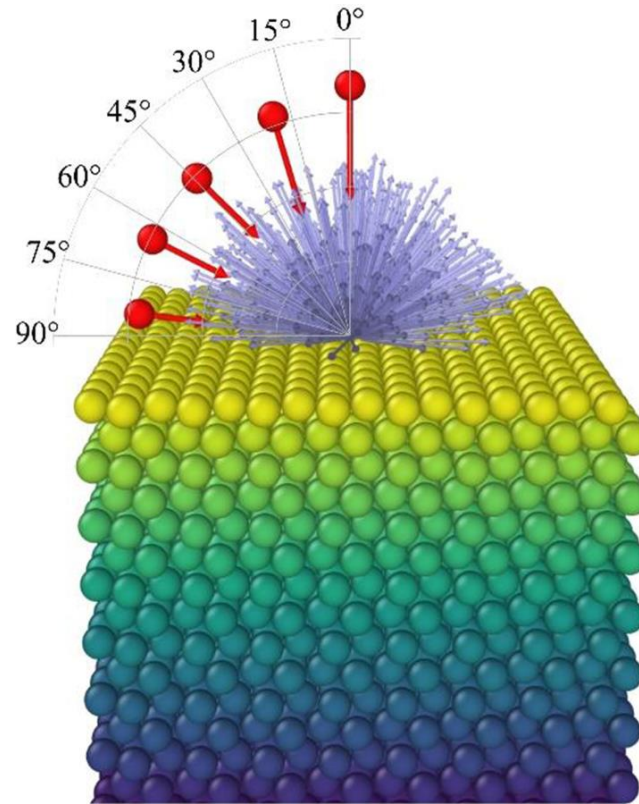
Sputtering

Implantation or Structural

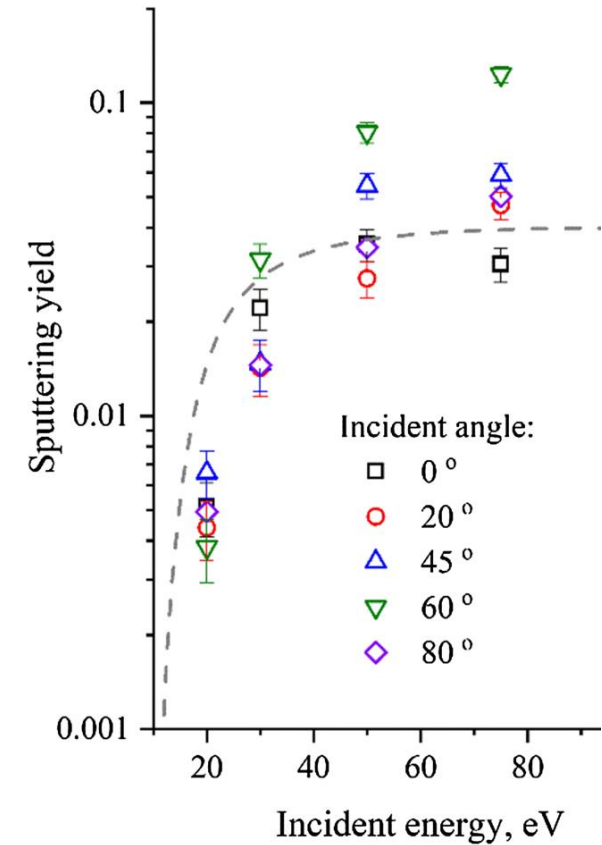


TARGET CONSIDERATIONS

- **Sputtering yield:**
 - Atoms ejected per incident ion.
 - Depends on
 - Target material
 - Ion energy
 - Ion mass
 - Incidence angle.



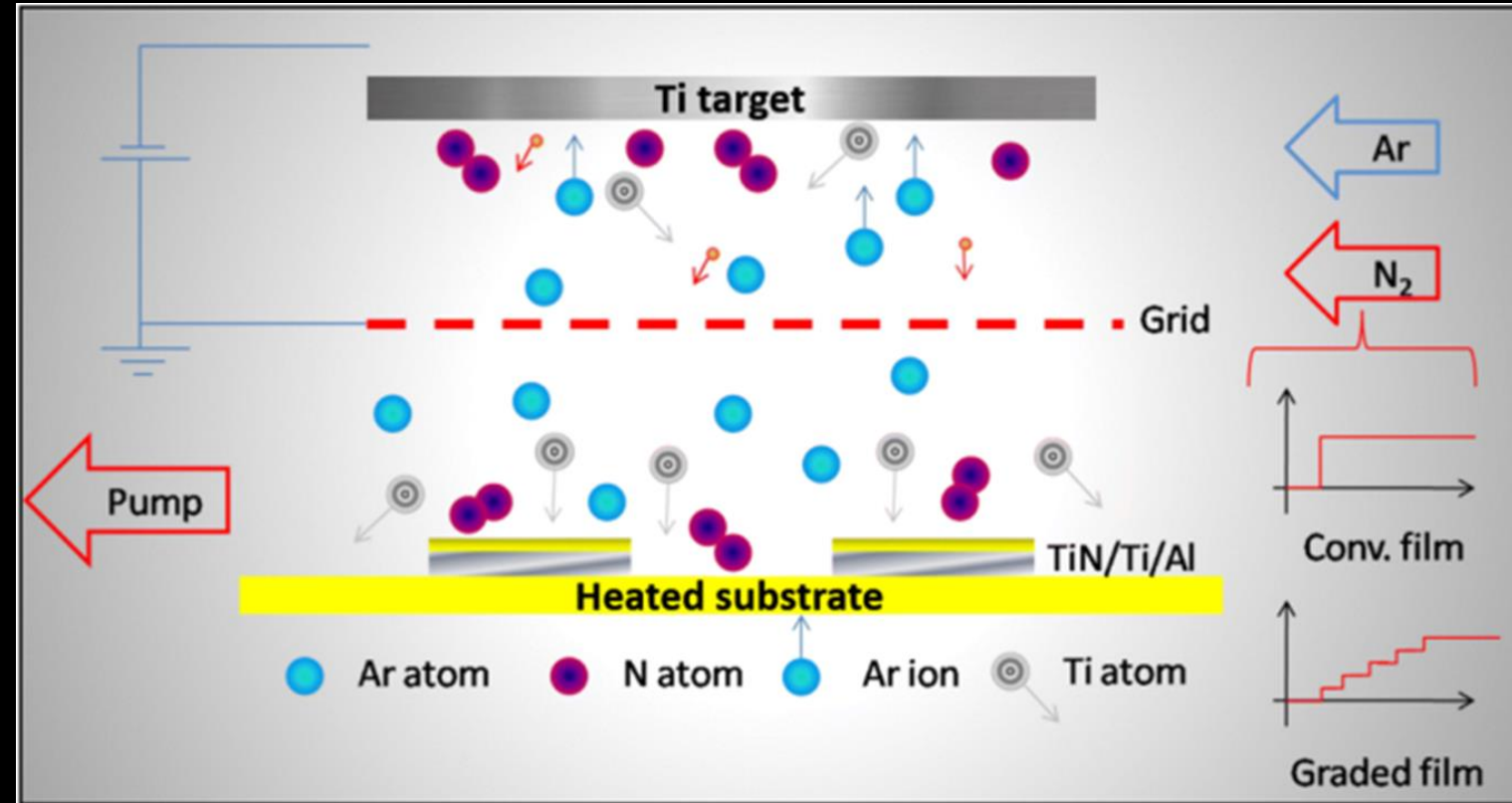
a



b

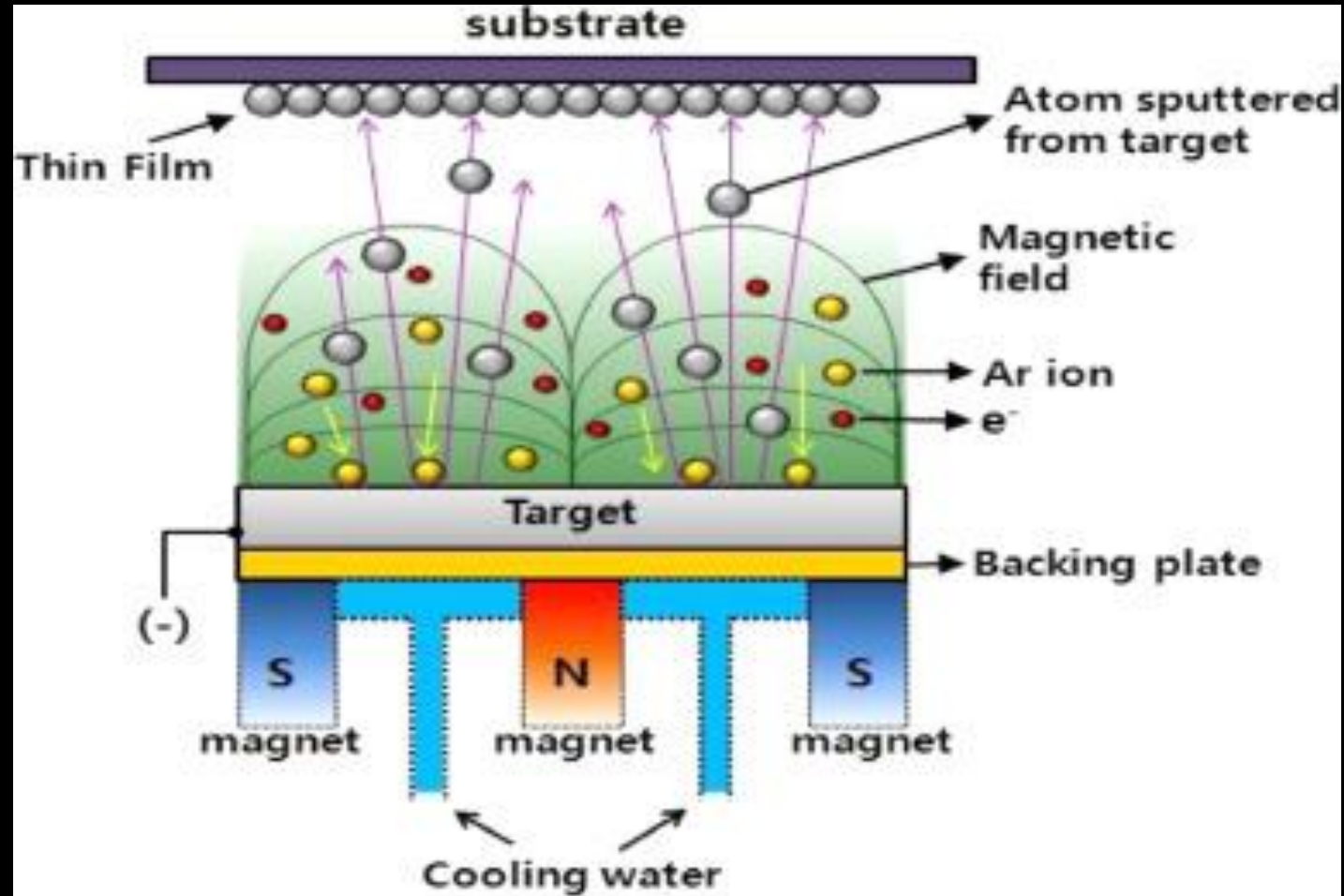
REACTIVE SPUTTERING

- Two modes:
 - **Compound-coated cathode:**
Reactant gas forms a compound on the target surface, which is sputtered onto the substrate.
 - **Metallic cathode:** Clean metallic target; compound forms only in the deposited material.



MAGNETRON SPUTTERING

- Plasma confined near the target using **strong magnetic fields**.
- Secondary electrons follow spiral trajectories, increasing ionization.
- Advantages:
 - High sputtering/deposition rates.
 - Low-impurity, high-quality films.



ADVANTAGES OF SPUTTERING

- **Versatility:** Metals, semiconductors, insulators, alloys, and compounds can be sputtered.
- **Low-temperature deposition:** Ideal for refractory materials.
- **Multicomponent films:** Ease of fabrication.
- **Uniformity:** Consistent thickness over large areas.
- **High adhesion:** Strong film-substrate bonds.

APPLICATIONS

- Microelectronics: **Insulating layers** and **thin-film resistors/capacitors**.
- Data storage: **Thin-film layers** in compact disks.
- Optoelectronics: **Transparent conducting electrodes** and **amorphous optical films**.
- Advanced materials: **Piezoelectric transducers** and **high-temperature superconductors**.
- Integrated circuits: Memory devices.